

Alex Liu

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Education

University of Florida (UF) - Gainesville, FL

Expected Graduation: **May 2028**

Bachelor of Science in Computer Science, Minor in Geography

GPA: 3.80/4.0

- **Relevant Coursework:** Computer Organization, Linear Algebra, Data Structures and Algorithms, Programming Fundamentals 2, Discrete Structures, Calculus 3

Skills

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL, RISC-V Assembly, Bash, HTML/CSS, GO, C#

Frameworks & Libraries: React.js, Node.js, FastAPI, Flask, TailwindCSS, SQLAlchemy, Pandas, NumPy

Developer Tools: Git, GitHub, Docker, GitHub Actions (CI/CD), Linux/Unix, VS Code, Postman, Vite

Platforms & Databases: PostgreSQL, Supabase, Vercel, Railway, Microservices, REST APIs, JWT

Experiences & Involvements

SASE Intern - UF Society of Asian Scientists & Engineers (SASE)

Aug 2025 - Dec 2025

- **Coordinated** professional events for **150+ attendees**, managing logistics and cross-functional team communication to enhance member career readiness.
- **Collaborated** with a leadership board of **100+** to streamline organizational programming and mentorship initiatives.

Event Committee Member - UF Chinese American Student Association (CASA)

Aug 2025 - Oct 2025

- **Spearheaded** a large-scale cultural festival for **300+ participants**, overseeing program design and day-of execution.
- **Curated** and managed cultural activities to increase student engagement and celebrate heritage for the UF community.

Instructor- Logic Lab

June 2022 – July 2022

- Led and instructed a class of **30+** students in math and chess, designing lessons with **4** other team members.

Projects

Geography Dashboard, *React, Leaflet, FastAPI, Vite, TypeScript, PostgreSQL, Docker, Gemini AI*

Nov 2025

- Engineered a **full-stack** geospatial analytics platform using a **microservices architecture**, decoupling core data processing from AI services to enable **distributed deployment** and independent service scaling.
- Architected a distributed backend infrastructure containerized with **Docker** and orchestrated via **CI/CD pipelines**, ensuring fault isolation and high availability across production environments.
- Integrated a cost-aware **AI** layer using Google Gemini, enforcing strict per-user rate limiting at the API layer to manage resource consumption across the distributed system.
- Developed high-performance **REST APIs** with **FastAPI** and **SQLAlchemy** to process structured geospatial datasets(CSV/GeoJSON), transforming raw spatial results into interpretive analytical summaries for city-scale data.

UF Health SmartScribe, *HTML, CSS, JavaScript, Supabase, Gemini AI*

Oct 2025

- Engineered a real-time **full-stack** medical documentation prototype within **24 hours**, integrating Deepgram and Gemini AI API to transcribe consultations and auto-generate structured SOAP notes.
- Architected a secure backend pipeline using **Supabase** to manage patient records, automating clinical workflows to reduce documentation time by **60%**.
- Implemented a speaker-aware AI extraction engine processing **60+ words/minute** with **sub-5-second** latency, prioritizing immediate physician utility over administrative overhead

Grade Track, *Flask, React.js, SQLAlchemy, PostgreSQL, Docker*

Oct 2025

- Engineered a full-stack analytics dashboard that visualizes student performance trends across courses using **Flask APIs, React, and Recharts** to integrate real-time data aggregation and visual analytics to demonstrate
- Orchestrated deployment in Docker, reducing **85%** setup time ensuring consistent cross-environment builds.

Ability Bridge, *Python, OpenCV, MediaPipe, PyAutoGUI*

Sep 2025

- Constructed a hands-free accessibility interface using OpenCV, processing **14,000+** facial landmarks/second to enable precise cursor control for users with limited mobility.
- Optimized signal processing algorithms (exponential smoothing) to achieve 30 FPS with sub-100ms latency, minimizing false activations across diverse lighting conditions.